Simulation Experiment 04

Code for MATLAB:

```
clc;
clear;
close all;

% Input specifications
Fs = 1000;
Fp = 100;
Fst = 150;
Rp = 1;
Rs = 40;

Wp = Fp/(Fs/2);
Ws = Fst/(Fs/2);

% 1. Butterworth Filter
[N1, Wn1] = buttord(Wp, Ws, Rp, Rs);
[b1, a1] = butter(N1, Wn1, 'low');

% 2. Chebyshev Type-I Filter
[N2, Wn2] = cheb1ord(Wp, Ws, Rp, Rs);
[b2, a2] = cheby1(N2, Rp, Wn2, 'low');

% 3. Elliptic Filter
[N3, Wn3] = ellipord(Wp, Ws, Rp, Rs);
[b3, a3] = ellip(N3, Rp, Rs, Wn3, 'low');

% Display filter orders
fprintf('FILTER COMPARISON RESULTS\n\n');
fprintf('Butterworth Order: %d\n', N1);
fprintf('Chebyshev Type-I Order: %d\n', N2);
fprintf('Elliptic Order: %d\n', N3);
fprintf('Transition Bandwidth: %.2f Hz\n', Fst-Fp);

% Frequency responses
[H1, f] = freqz(b1, a1, 2048, Fs);
H2 = freqz(b2, a2, 2048, Fs);
H3 = freqz(b3, a3, 2048, Fs);

% Magnitude response comparison
figure;
plot(f, 20*log10(max(abs(H1),1e-10)), 'b', ...
     f, 20*log10(max(abs(H2),1e-10)), 'r--', ...
     f, 20*log10(max(abs(H3),1e-10)), 'g-.', ...
     'LineWidth', 1.3);
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Butterworth vs Chebyshev-I vs Elliptic');
legend('Butterworth','Chebyshev Type-I','Elliptic');
xline(Fp, ':k', 'Passband Edge');
xline(Fst, ':m', 'Stopband Edge');

% Phase response comparison
figure;
plot(f, unwrap(angle(H1))*180/pi, 'b', ...
     f, unwrap(angle(H2))*180/pi, 'r--', ...
     f, unwrap(angle(H3))*180/pi, 'g-.', ...
```

```
     'LineWidth', 1.3);
grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Phase Response Comparison');
legend('Butterworth','Chebyshev Type-I','Elliptic');

% Display coefficients
disp('Butterworth Coefficients:');
disp('Numerator:'); disp(b1);
disp('Denominator:'); disp(a1);

disp('Chebyshev Type-I Coefficients:');
disp('Numerator:'); disp(b2);
disp('Denominator:'); disp(a2);

disp('Elliptic Coefficients:');
disp('Numerator:'); disp(b3);
disp('Denominator:'); disp(a3);
```

Command window results:

Filter Order Comparison

- Butterworth Filter Order: 12
- Chebyshev Type-I Filter Order: 6
- Elliptic Filter Order: 4
- Transition Bandwidth: 50 Hz

Filter Coefficients

1. Butterworth Filter

- Numerator (b):

```
[0.0000002, 0.0000027, 0.0000146, 0.0000488,
 0.0001098, 0.0001757, 0.0002050, 0.0001757,
 0.0001098, 0.0000488, 0.0000146, 0.0000027,
 0.0000002]
```

- Denominator (a):

```
[1, -6.8844, 22.4341, -45.5100, 63.7725,
 -64.8501, 48.9643, -27.6092, 11.5218,
 -3.4662, 0.7128, -0.0899, 0.0053]
```

2. Chebyshev Type-I Filter

- Numerator (b):

```
[0.0000464, 0.0002782, 0.0006956, 0.0009274,
 0.0006956, 0.0002782, 0.0000464]
```

- Denominator (a):

```
[1, -4.8694, 10.3807, -12.3367, 8.5969,
 -3.3268, 0.5586]
```

3. Elliptic Filter

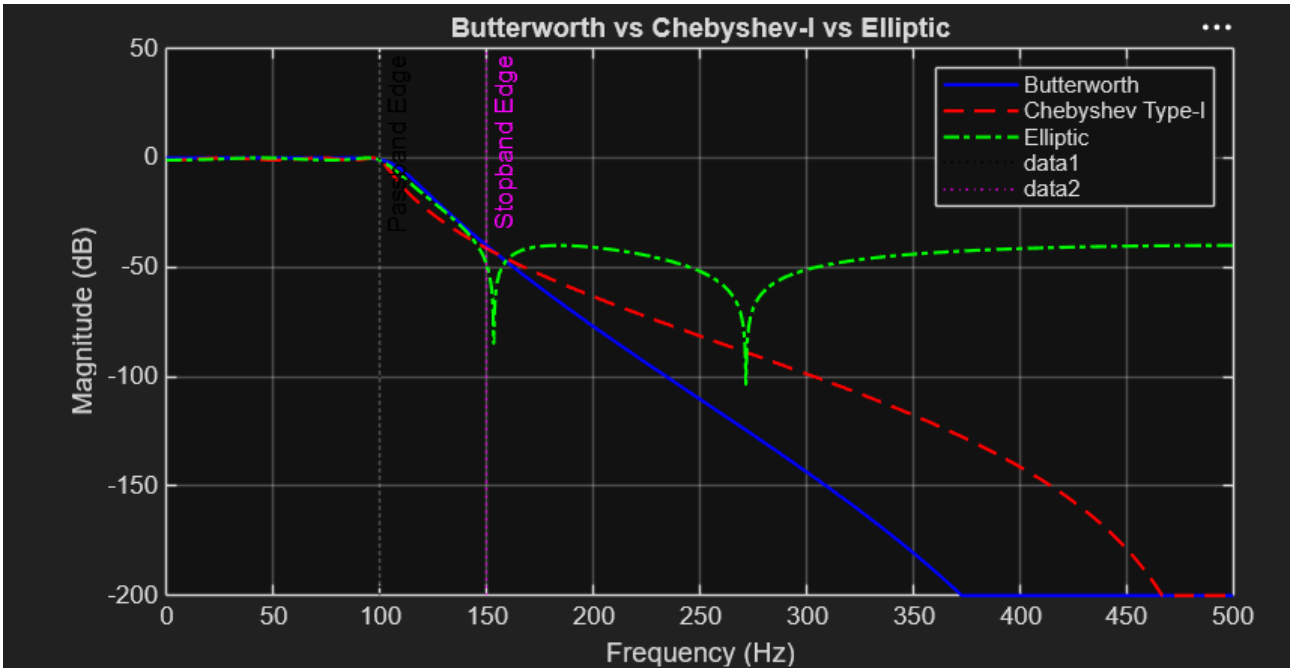
- Numerator (b):

```
[0.0197, -0.0171, 0.0333, -0.0171, 0.0197]
```

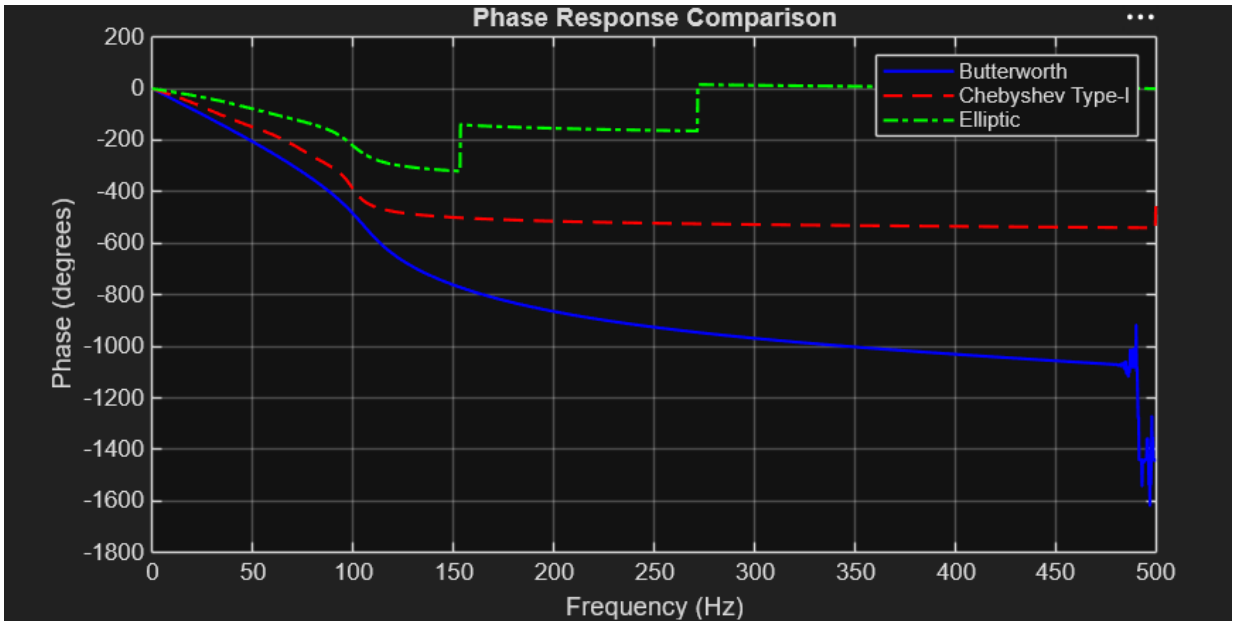
- Denominator (a):

```
[1, -3.0330, 3.8118, -2.2911, 0.5554]
```

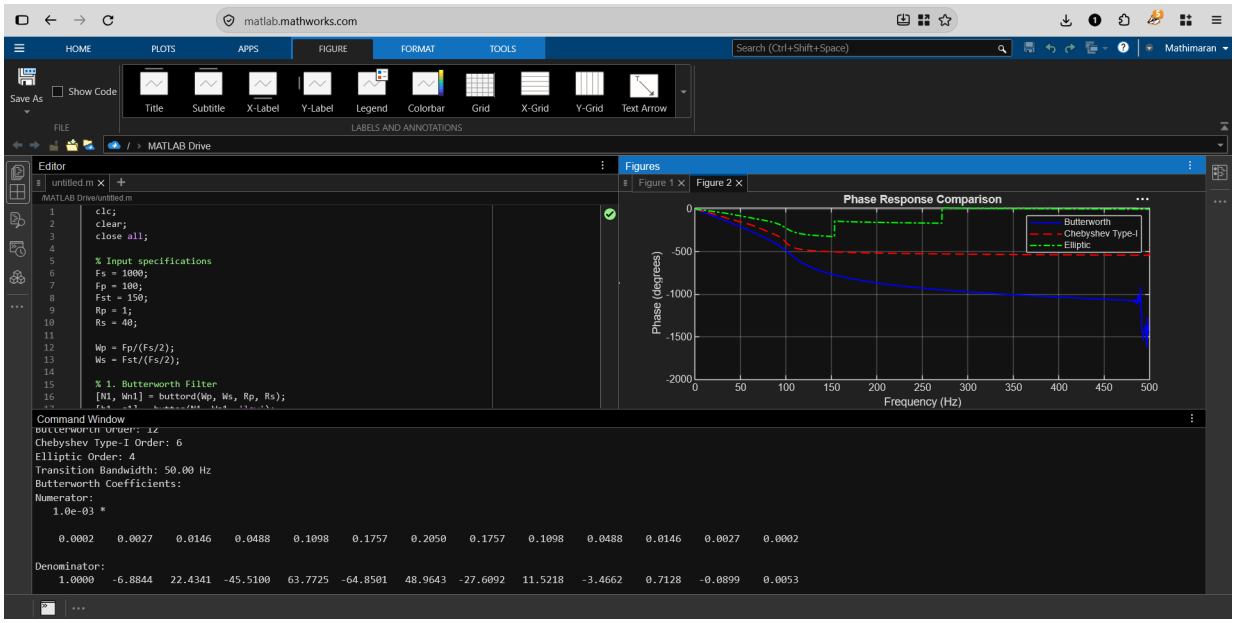
Graphs:
1. Butterworth vs Chebyshev-I vs Elliptic




2. Phase response graph



Screenshot:




Objectives:

1. To design Butterworth, Chebyshev Type-I, and Elliptic IIR filters using MATLAB.
2. To determine and compare the filter orders for identical frequency specifications.
3. To analyze the passband ripple, stopband attenuation, and transition bandwidth.
4. To plot and compare the magnitude and phase responses of the three filters.
5. To evaluate the trade-offs between filter order, ripple, transition sharpness, and filter complexity.

---------All the Objectives are simulated----------